

Two-Sample Tests of Hypothesis

11



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- ▲ **GIBBS BABY FOOD COMPANY** wishes to compare the weight gain of infants using its brand versus its competitor's. A sample of 40 babies using the Gibbs products revealed a mean weight gain of 7.6 pounds in the first three months after birth. For the Gibbs brand, the population standard deviation of the sample is 2.3 pounds. A sample of 55 babies using the competitor's brand revealed a mean increase in weight of 8.1 pounds. The population standard deviation is 2.9 pounds. At the .05 significance level, can we conclude that babies using the Gibbs brand gained less weight? (See Exercise 3 and **LO11-1**.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

- LO11-1** Test a hypothesis that two independent population means are equal, assuming that the population standard deviations are known and equal.
- LO11-2** Test a hypothesis that two independent population means are equal, with unknown population standard deviations.
- LO11-3** Test a hypothesis about the mean population difference between paired or dependent observations.
- LO11-4** Explain the difference between dependent and independent samples.

INTRODUCTION

Chapter 10 began our study of hypothesis testing. We described the nature of hypothesis testing and conducted tests of a hypothesis in which we compared the results of a single sample to a population value. That is, we selected a single random sample from a population and conducted a test of whether the proposed population value was reasonable. Recall in Chapter 10 that we selected a sample of the number of desks assembled per week at Jamestown Steel Company to determine whether there was a change

in the production rate. Similarly, we sampled the cost to process insurance claims to determine if cost-cutting measures resulted in a mean less than the current \$60 per claim. In both cases, we compared the results of a *single* sample statistic to a population parameter.

In this chapter, we expand the idea of hypothesis testing to two populations. That is, we select random samples from two different populations to determine whether the population means are equal. Some questions we might want to test are:



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1. Is there a difference in the mean value of residential real estate sold by male agents and female agents in south Florida?
2. At Grabit Software, Inc., do customer service employees receive more calls for assistance during the morning or afternoon?
3. In the fast-food industry, is there a difference in the mean number of days absent between young workers (under 21 years of age) and older workers (more than 60 years of age)?
4. Is there an increase in the production rate if music is piped into the production area?

We begin this chapter with the case in which we select random samples from two independent populations and wish to investigate whether these populations have the same mean.

LO11-1

Test a hypothesis that two independent population means are equal, assuming that the population standard deviations are known and equal.

TWO-SAMPLE TESTS OF HYPOTHESIS: INDEPENDENT SAMPLES

A city planner in Tampa, Florida wishes to know whether there is a difference in the mean hourly wage rate of plumbers and electricians in central Florida. A financial accountant wishes to know whether the mean rate of return for domestic, U.S., mutual funds is different from the mean rate of return on global mutual funds. In each of these cases, there are two independent populations. In the first case, the plumbers represent one population and the electricians, the other. In the second case, domestic, U.S., mutual funds are one population and global mutual funds, the other.

To investigate the question in each of these cases, we would select a random sample from each population and compute the mean of the two samples. If the two population means are the same, that is, the mean hourly rate is the same for the plumbers and the electricians, we would expect the *difference* between the two sample means to be zero. But what if our sample results yield a difference other than zero? Is that difference due to chance or is it because there is a real difference in the hourly earnings? A two-sample test of means will help to answer this question.

Return to the results of Chapter 8. Recall that we showed that a distribution of sample means would tend to approximate the normal distribution. We need to again assume that a distribution of sample means will follow the normal distribution. It can be shown

mathematically that the distribution of the differences between sample means for two normal distributions is also normal.

We can illustrate this theory in terms of the city planner in Tampa, Florida. To begin, let's assume some information that is not usually available. Suppose that the population of plumbers has a mean of \$30.00 per hour and a standard deviation of \$5.00 per hour. The population of electricians has a mean of \$29.00 and a standard deviation of \$4.50. Now, from this information it is clear that the two population means are not the same. The plumbers actually earn \$1.00 per hour more than the electricians. But we cannot expect to uncover this difference each time we sample the two populations.

Suppose we select a random sample of 40 plumbers and a random sample of 35 electricians and compute the mean of each sample. Then, we determine the difference between the sample means. It is this difference between the sample means that holds our interest. If the populations have the same mean, then we would expect the difference between the two sample means to be zero. If there is a difference between the population means, then we expect to find a difference between the sample means.

To understand the theory, we need to take several pairs of samples, compute the mean of each, determine the difference between the sample means, and study the distribution of the differences in the sample means. Because of the Central Limit Theorem in Chapter 8, we know that the distribution of the sample means follows the normal distribution. If the two distributions of sample means follow the normal distribution, then we can reason that the distribution of their differences will also follow the normal distribution. This is the first hurdle.

The second hurdle refers to the mean of this distribution of differences. If we find the mean of this distribution is zero, that implies that there is no difference in the two populations. On the other hand, if the mean of the distribution of differences is equal to some value other than zero, either positive or negative, then we conclude that the two populations do not have the same mean.

To report some concrete results, let's return to the city planner in Tampa, Florida. Table 11–1 shows the result of selecting 20 different samples of 40 plumbers and 35 electricians, computing the mean of each sample, and finding the difference

TABLE 11–1 The Mean Hourly Earnings of 20 Random Samples of Plumbers and Electricians and the Differences between the Means

Sample	Plumbers	Electricians	Difference
1	\$29.80	\$28.76	\$1.04
2	30.32	29.40	0.92
3	30.57	29.94	0.63
4	30.04	28.93	1.11
5	30.09	29.78	0.31
6	30.02	28.66	1.36
7	29.60	29.13	0.47
8	29.63	29.42	0.21
9	30.17	29.29	0.88
10	30.81	29.75	1.06
11	30.09	28.05	2.04
12	29.35	29.07	0.28
13	29.42	28.79	0.63
14	29.78	29.54	0.24
15	29.60	29.60	0.00
16	30.60	30.19	0.41
17	30.79	28.65	2.14
18	29.14	29.95	−0.81
19	29.91	28.75	1.16
20	28.74	29.21	−0.47

between the two sample means. In the first case, the sample of 40 plumbers has a mean of \$29.80, and for the 35 electricians the mean is \$28.76. The difference between the sample means is \$1.04. This process was repeated 19 more times. Observe that in 17 of the 20 cases, the differences are positive because the mean of the plumbers is larger than the mean of the electricians. In two cases, the differences are negative because the mean of the electricians is larger than the mean of the plumbers. In one case, the means are equal.

Our final hurdle is that we need to know something about the *variability* of the distribution of differences. To put it another way, what is the standard deviation of this distribution of differences? Statistical theory shows that when we have independent populations, as in this case, the distribution of the differences has a variance (standard deviation squared) equal to the sum of the two individual variances. This means that we can add the variances of the two sampling distributions. To put it another way, the variance of the difference in sample means ($\bar{x}_1 - \bar{x}_2$) is equal to the sum of the variance for the plumbers and the variance for the electricians.

**VARIANCE OF THE DISTRIBUTION
OF DIFFERENCES IN MEANS**

$$\sigma_{\bar{x}_1 - \bar{x}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \quad (11-1)$$

The term $\sigma_{\bar{x}_1 - \bar{x}_2}^2$ looks complex but need not be difficult to interpret. The σ^2 portion reminds us that it is a variance, and the subscript $\bar{x}_1 - \bar{x}_2$ that it is a distribution of differences in the sample means.

We can put this equation in a more usable form by taking the square root, so that we have the standard deviation or “standard error” of the distribution of differences. Finally, we standardize the distribution of the differences. The result is the following equation.

**TWO-SAMPLE TEST OF
MEANS—KNOWN σ**

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (11-2)$$

Before we present an example, let’s review the assumptions necessary for using formula (11-2).

- The two populations follow normal distributions.
- The two samples are unrelated, that is, independent.
- The standard deviations for both populations are known.

The following example shows the details of the test of hypothesis for two population means and shows how to interpret the results.

► **EXAMPLE**

Customers at the FoodTown Supermarket have a choice when paying for their groceries. They may check out and pay using the standard cashier-assisted checkout, or they may use the new Fast Lane procedure. In the standard procedure, a FoodTown employee scans each item and puts it on a short conveyor, where another employee puts it in a bag and then into the grocery cart. In the Fast Lane procedure,



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the customer scans each item, bags it, and places the bags in the cart him- or herself. The Fast Lane procedure is designed to reduce the time a customer spends in the checkout line.

The Fast Lane facility was recently installed at the Byrne Road FoodTown location. The store manager would like to know if the mean checkout time using the standard checkout method is longer than using the Fast Lane. She gathered the following sample information. The time is measured from when the customer enters the line until all his or her bags are in the cart. Hence the time includes both waiting in line and checking out. What is the p -value?

Customer Type	Sample Size	Sample Mean	Population Standard Deviation
Standard	50	5.50 minutes	0.40 minute
Fast Lane	100	5.30 minutes	0.30 minute

SOLUTION

We use the six-step hypothesis-testing procedure to investigate the question.

Step 1: State the null hypothesis and the alternate hypothesis. The null hypothesis is that the mean standard checkout time is less than or equal to the mean Fast Lane checkout time. In other words, the difference of 0.20 minute between the mean checkout time for the standard method and the mean checkout time for Fast Lane is due to chance. The alternate hypothesis is that the mean checkout time is larger for those using the standard method. We will let μ_S refer to the mean checkout time for the population of standard customers and μ_F the mean checkout time for the Fast Lane customers. The null and alternative hypotheses are:

$$H_0: \mu_S \leq \mu_F$$

$$H_1: \mu_S > \mu_F$$

Step 2: Select the level of significance. The significance level is the probability that we reject the null hypothesis when it is actually true. This likelihood is determined prior to selecting the sample or performing any calculations. The .05 and .01 significance levels are the most common, but other values, such as .02 and .10, are also used. In theory, we may select any value between 0 and 1 for the significance level. In this case, we selected the .01 significance level.

Step 3: Determine the test statistic. In Chapter 10, we used the standard normal distribution (that is, z) and t as test statistics. In this case, we use the z distribution as the test statistic because we assume the two population distributions are both normal and the standard deviations of both populations are known.

Step 4: Formulate a decision rule. The decision rule is based on the null and the alternate hypotheses (i.e., one-tailed or two-tailed test), the level of significance, and the test statistic used. We selected the .01 significance level and the z distribution as the test statistic, and we wish to determine whether the mean checkout time is longer using the standard method. We set the alternate hypothesis to indicate that the mean checkout time is longer for those using the standard method than the Fast Lane method. Hence, the rejection region is in the upper tail of the standard normal distribution (a one-tailed test). To find the critical value, go to Student's t distribution

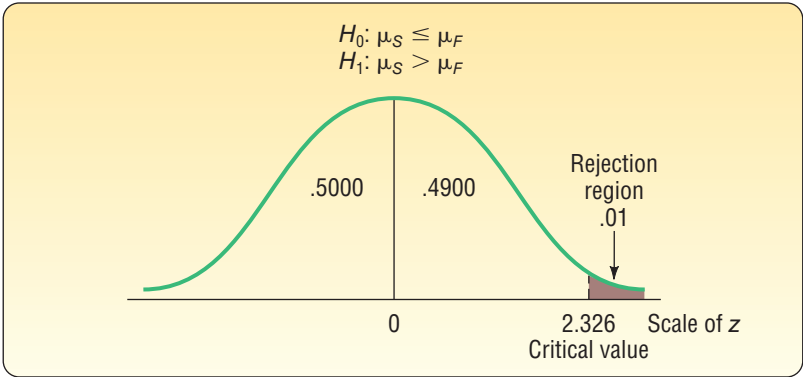


CHART 11–1 Decision Rule for One-Tailed Test at .01 Significance Level

(Appendix B.5). In the table headings, find the row labeled “**Level of Significance for One-Tailed Test**” and select the column for an alpha of .01. Go to the bottom row with infinite degrees of freedom. The z critical value is 2.326. So the decision rule is to reject the null hypothesis if the value of the test statistic exceeds 2.326. Chart 11–1 depicts the decision rule.

Step 5: Make the decision regarding H_0 . FoodTown randomly selected 50 customers using the standard checkout and computed a sample mean checkout time of 5.5 minutes, and 100 customers using the Fast Lane checkout and computed a sample mean checkout time of 5.3 minutes. We assume that the population standard deviations for the two methods is known. We use formula (11-2) to compute the value of the test statistic.

$$z = \frac{\bar{x}_S - \bar{x}_F}{\sqrt{\frac{\sigma_S^2}{n_S} + \frac{\sigma_F^2}{n_F}}} = \frac{5.5 - 5.3}{\sqrt{\frac{0.40^2}{50} + \frac{0.30^2}{100}}} = \frac{0.2}{0.064031} = 3.123$$

The computed value of 3.123 is larger than the critical value of 2.326. Our decision is to reject the null hypothesis and accept the alternate hypothesis.

Step 6: Interpret the result. The difference of .20 minute between the mean checkout times is too large to have occurred by chance. We conclude the Fast Lane method is faster.

What is the p -value for the test statistic? Recall that the p -value is the probability of finding a value of the test statistic this extreme when the null hypothesis is true. To calculate the p -value, we need the probability of a z value larger than 3.123. From Appendix B.3, we cannot find the probability associated with 3.123. The largest value available is 3.09. The area corresponding to 3.09 is .4990. In this case, we can report that the p -value is less than .0010, found by .5000 – .4990. We conclude that there is very little likelihood that the null hypothesis is true! The checkout time is less using the fast lane.

STATISTICS IN ACTION

Do you live to work or work to live? A recent poll of 802 working Americans revealed that, among those who considered their work as a career, the mean number of hours worked per day was 8.7. Among those who considered their work as a job, the mean number of hours worked per day was 7.6.

In summary, the criteria for using formula (11–2) are:

- 1. **The samples are from independent populations.** This means the checkout time for the Fast Lane customers is unrelated to the checkout time for the other customers. For example, Mr. Smith’s checkout time does not affect any other customer’s checkout time.

2. **Both populations follow the normal distribution.** In the FoodTown example, the population of times in both the standard checkout line and the Fast Lane follow normal distributions.
3. **Both population standard deviations are known.** In the FoodTown example, the population standard deviation of the Fast Lane times was 0.30 minute. The population standard deviation of the standard checkout times was 0.40 minute.

SELF-REVIEW 11-1



Tom Sevits is the owner of the Appliance Patch. Recently Tom observed a difference in the dollar value of sales between the men and women he employs as sales associates. A sample of 40 days revealed the men sold a mean of \$1,400 worth of appliances per day. For a sample of 50 days, the women sold a mean of \$1,500 worth of appliances per day. Assume the population standard deviation for men is \$200 and for women \$250. At the .05 significance level, can Mr. Sevits conclude that the mean amount sold per day is larger for the women?

- (a) State the null hypothesis and the alternate hypothesis.
- (b) What is the decision rule?
- (c) What is the value of the test statistic?
- (d) What is your decision regarding the null hypothesis?
- (e) What is the p -value?
- (f) Interpret the result.

EXERCISES

1. A sample of 40 observations is selected from one population with a population standard deviation of 5. The sample mean is 102. A sample of 50 observations is selected from a second population with a population standard deviation of 6. The sample mean is 99. Conduct the following test of hypothesis using the .04 significance level.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

- a. Is this a one-tailed or a two-tailed test?
 - b. State the decision rule.
 - c. Compute the value of the test statistic.
 - d. What is your decision regarding H_0 ?
 - e. What is the p -value?
2. A sample of 65 observations is selected from one population with a population standard deviation of 0.75. The sample mean is 2.67. A sample of 50 observations is selected from a second population with a population standard deviation of 0.66. The sample mean is 2.59. Conduct the following test of hypothesis using the .08 significance level.

$$H_0: \mu_1 \leq \mu_2$$

$$H_1: \mu_1 > \mu_2$$

- a. Is this a one-tailed or a two-tailed test?
- b. State the decision rule.
- c. Compute the value of the test statistic.
- d. What is your decision regarding H_0 ?
- e. What is the p -value?

Note: Use the six-step hypothesis-testing procedure to solve the following exercises.

3. Gibbs Baby Food Company wishes to compare the weight gain of infants using its brand versus its competitor's. A sample of 40 babies using the Gibbs products revealed a mean weight gain of 7.6 pounds in the first three months after birth. For the Gibbs brand, the population standard deviation of the sample is 2.3 pounds. A

sample of 55 babies using the competitor’s brand revealed a mean increase in weight of 8.1 pounds. The population standard deviation is 2.9 pounds. At the .05 significance level, can we conclude that babies using the Gibbs brand gained less weight? Compute the p -value and interpret it.

- 4. As part of a study of corporate employees, the director of human resources for PNC Inc. wants to compare the distance traveled to work by employees at its office in downtown Cincinnati with the distance for those in downtown Pittsburgh. A sample of 35 Cincinnati employees showed they travel a mean of 370 miles per month. A sample of 40 Pittsburgh employees showed they travel a mean of 380 miles per month. The population standard deviations for the Cincinnati and Pittsburgh employees are 30 and 26 miles, respectively. At the .05 significance level, is there a difference in the mean number of miles traveled per month between Cincinnati and Pittsburgh employees?
- 5. Do married and unmarried women spend the same amount of time per week using Facebook? A random sample of 45 married women who use Facebook spent an average of 3.0 hours per week on this social media website. A random sample of 39 unmarried women who regularly use Facebook spent an average of 3.4 hours per week. Assume that the weekly Facebook time for married women has a population standard deviation of 1.2 hours, and the population standard deviation for unmarried, regular Facebook users is 1.1 hours per week. Using the .05 significance level, do married and unmarried women differ in the amount of time per week spent on Facebook? Find the p -value and interpret the result.
- 6. Mary Jo Fitzpatrick is the vice president for Nursing Services at St. Luke’s Memorial Hospital. Recently she noticed in the job postings for nurses that those that are unionized seem to offer higher wages. She decided to investigate and gathered the following information.

Group	Sample Size	Sample Mean Wage	Population Standard Deviation
Union	40	\$20.75	\$2.25
Nonunion	45	\$19.80	\$1.90

Would it be reasonable for her to conclude that union nurses earn more? Use the .02 significance level. What is the p -value?

LO11-2

Test a hypothesis that two independent population means are equal, with unknown population standard deviations.

COMPARING POPULATION MEANS WITH UNKNOWN POPULATION STANDARD DEVIATIONS

In the previous section, we used the standard normal distribution and z as the test statistic to test a hypothesis that two population means from independent populations were equal. The hypothesis tests presumed that the populations were normally distributed and that we knew the population standard deviations. However, in most cases, we do not know the population standard deviations. We can overcome this problem, as we did in the one-sample case in the previous chapter, by substituting the sample standard deviation (s) for the population standard deviation (σ). See formula (10–2) on page 334.

Two-Sample Pooled Test

In this section, we describe another method for comparing the sample means of two independent populations to determine if the sampled populations could reasonably have the same mean. The method described does *not* require that we know the standard deviations of the populations. This gives us a great deal more flexibility when

investigating the difference in sample means. There are two major differences in this test and the previous test described in this chapter.

1. We assume the sampled populations have equal but unknown standard deviations. Because of this assumption, we combine or “pool” the sample standard deviations.
2. We use the t distribution as the test statistic.

The formula for computing the value of the test statistic t is similar to formula (11-2), but an additional calculation is necessary. The two sample standard deviations are pooled to form a single estimate of the unknown population standard deviation. In essence, we compute a weighted mean of the two sample standard deviations and use this value as an estimate of the unknown population standard deviation. The weights are the degrees of freedom that each sample provides. Why do we need to pool the sample standard deviations? Because we assume that the two populations have equal standard deviations, the best estimate we can make of that value is to combine or pool all the sample information we have about the value of the population standard deviation.

The following formula is used to pool the sample standard deviations. Notice that two factors are involved: the number of observations in each sample and the sample standard deviations themselves.

POOLED VARIANCE

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \quad (11-3)$$

where:

- s_1^2 is the variance (standard deviation squared) of the first sample.
- s_2^2 is the variance of the second sample.

The value of t is computed from the following equation.

**TWO-SAMPLE TEST OF MEANS—
UNKNOWN σ 'S**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (11-4)$$

where:

- $\bar{x}_1$ is the mean of the first sample.
- $\bar{x}_2$ is the mean of the second sample.
- n_1 is the number of observations in the first sample.
- n_2 is the number of observations in the second sample.
- s_p^2 is the pooled estimate of the population variance.

The number of degrees of freedom in the test is the total number of items sampled minus the total number of samples. Because there are two samples, there are $n_1 + n_2 - 2$ degrees of freedom.

To summarize, there are three requirements or assumptions for the test.

1. The sampled populations are approximately normally distributed.
2. The sampled populations are independent.
3. The standard deviations of the two populations are equal.

The following example/solution explains the details of the test.

 EXAMPLE

Owens Lawn Care Inc. manufactures and assembles lawnmowers that are shipped to dealers throughout the United States and Canada. Two different procedures have been proposed for mounting the engine on the frame of the lawnmower. The question is: Is there a difference in the mean time to mount the engines on the

frames of the lawnmowers? The first procedure was developed by longtime Owens employee Herb Welles (designated as procedure W), and the other procedure was developed by Owens Vice President of Engineering William Atkins (designated as procedure A). To evaluate the two methods, we conduct a time and motion study. A sample of five employees is timed using the Welles method and six using the Atkins method. The results, in minutes, are shown below. Is there a difference in the mean mounting times? Use the .10 significance level.

Welles (minutes)	Atkins (minutes)
2	3
4	7
9	5
3	8
2	4
	3

SOLUTION

Following the six steps to test a hypothesis, the null hypothesis states that there is no difference in mean mounting times between the two procedures. The alternate hypothesis indicates that there is a difference.

$H_0: \mu_W = \mu_A$
 $H_1: \mu_W \neq \mu_A$

The required assumptions are:

- The observations in the Welles sample are *independent* of the observations in the Atkins sample.
- The two populations follow the normal distribution.
- The two populations have equal standard deviations.

Is there a difference between the mean assembly times using the Welles and the Atkins methods? The degrees of freedom are equal to the total number of items sampled minus the number of samples. In this case, that is $n_W + n_A - 2$. Five assemblers used the Welles method and six the Atkins method. Thus, there are 9 degrees of freedom, found by $5 + 6 - 2$. The critical values of t , from Appendix B.5 for $df = 9$, a two-tailed test, and the .10 significance level, are -1.833 and 1.833 . The decision rule is portrayed graphically in Chart 11–2. We do not reject the null hypothesis if the computed value of t falls between -1.833 and 1.833 .

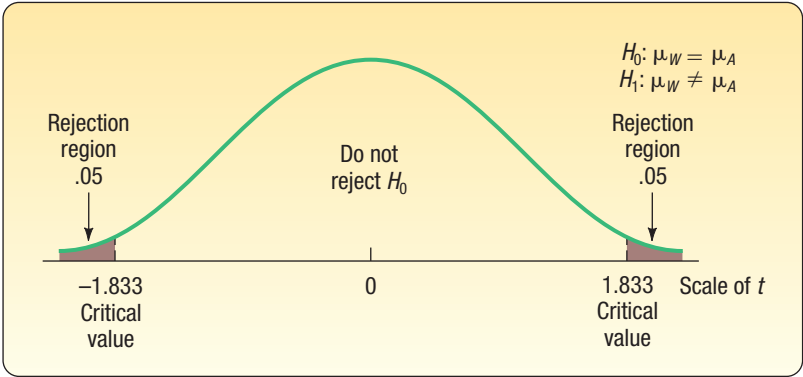


CHART 11–2 Regions of Rejection, Two-Tailed Test, $df = 9$, and .10 Significance Level

We use three steps to compute the value of t .

Step 1: Calculate the sample standard deviations. To compute the sample standard deviations, we use formula (3–9). See the details below.

Welles Method		Atkins Method	
x_W	$(x_W - \bar{x}_W)^2$	x_A	$(x_A - \bar{x}_A)^2$
2	$(2 - 4)^2 = 4$	3	$(3 - 5)^2 = 4$
4	$(4 - 4)^2 = 0$	7	$(7 - 5)^2 = 4$
9	$(9 - 4)^2 = 25$	5	$(5 - 5)^2 = 0$
3	$(3 - 4)^2 = 1$	8	$(8 - 5)^2 = 9$
2	$(2 - 4)^2 = 4$	4	$(4 - 5)^2 = 1$
20	34	3	$(3 - 5)^2 = 4$
		30	22

$$\bar{x}_W = \frac{\Sigma x_W}{n_W} = \frac{20}{5} = 4 \qquad \bar{x}_A = \frac{\Sigma x_A}{n_A} = \frac{30}{6} = 5$$

$$s_W = \sqrt{\frac{\Sigma(x_W - \bar{x}_W)^2}{n_W - 1}} = \sqrt{\frac{34}{5 - 1}} = 2.9155 \qquad s_A = \sqrt{\frac{\Sigma(x_A - \bar{x}_A)^2}{n_A - 1}} = \sqrt{\frac{22}{6 - 1}} = 2.0976$$

Step 2: Pool the sample variances. We use formula (11–3) to pool the sample variances (standard deviations squared).

$$s_p^2 = \frac{(n_W - 1)s_W^2 + (n_A - 1)s_A^2}{n_W + n_A - 2} = \frac{(5 - 1)(2.9155)^2 + (6 - 1)(2.0976)^2}{5 + 6 - 2} = 6.2222$$

Step 3: Determine the value of t . The mean mounting time for the Welles method is 4.00 minutes, found by $\bar{x}_W = 20/5$. The mean mounting time for the Atkins method is 5.00 minutes, found by $\bar{x}_A = 30/6$. We use formula (11–4) to calculate the value of t .

$$t = \frac{\bar{x}_W - \bar{x}_A}{\sqrt{s_p^2 \left(\frac{1}{n_W} + \frac{1}{n_A} \right)}} = \frac{4.00 - 5.00}{\sqrt{6.2222 \left(\frac{1}{5} + \frac{1}{6} \right)}} = -0.662$$

The decision is not to reject the null hypothesis because -0.662 falls in the region between -1.833 and 1.833 . Our conclusion is that the sample data failed to show a difference between the mean assembly times of the two methods.

We also can estimate the p -value using Appendix B.5. Locate the row with 9 degrees of freedom, and use the two-tailed test column. Find the t value, without regard to the sign, that is closest to our computed value of 0.662 . It is 1.383 , corresponding to a significance level of $.20$. Thus, even had we used the 20% significance level, we would not have rejected the null hypothesis of equal means. We can report that the p -value is greater than $.20$.

Excel has a procedure called “t-Test: Two Sample Assuming Equal Variances” that will perform the calculations of formulas (11–3) and (11–4) as well as find the sample means and sample variances. The details of the procedure are provided in Appendix C. The data are input in the first two columns of the Excel spreadsheet. They are labeled “Welles” and “Atkins.” The output follows. The value of t , called the “t Stat,” is -0.662 , and the two-tailed p -value is $.525$. As we would expect, the p -value is larger than the significance level of $.10$. The conclusion is not to reject the null hypothesis.

	A	B	C	D	E	F
1	Welles	Atkins		t-Test: Two-Sample Assuming Equal Variances		
2	2	3				
3	4	7			Welles	Atkins
4	9	5		Mean	4.000	5.000
5	3	8		Variance	8.500	4.400
6	2	4		Observations	5.000	6.000
7		3		Pooled Variance	6.222	
8				Hypothesized Mean Difference	0.000	
9				df	9.000	
10				t Stat	-0.662	
11				P(T<=t) one-tail	0.262	
12				t Critical one-tail	1.833	
13				P(T<=t) two-tail	0.525	
14				t Critical two-tail	2.262	

SELF-REVIEW 11-2



The production manager at Bellevue Steel, a manufacturer of wheelchairs, wants to compare the number of defective wheelchairs produced on the day shift with the number on the afternoon shift. A sample of the production from 6 day shifts and 8 afternoon shifts revealed the following number of defects.

Day	5	8	7	6	9	7		
Afternoon	8	10	7	11	9	12	14	9

- At the .05 significance level, is there a difference in the mean number of defects per shift?
- (a) State the null hypothesis and the alternate hypothesis.
 - (b) What is the decision rule?
 - (c) What is the value of the test statistic?
 - (d) What is your decision regarding the null hypothesis?
 - (e) What is the *p*-value?
 - (f) Interpret the result.
 - (g) What are the assumptions necessary for this test?

EXERCISES

For Exercises 7 and 8: (a) state the decision rule, (b) compute the pooled estimate of the population variance, (c) compute the test statistic, (d) state your decision about the null hypothesis, and (e) estimate the *p*-value.

7. The null and alternate hypotheses are:
- $$H_0: \mu_1 = \mu_2$$
$$H_1: \mu_1 \neq \mu_2$$

A random sample of 10 observations from one population revealed a sample mean of 23 and a sample standard deviation of 4. A random sample of 8 observations from another population revealed a sample mean of 26 and a sample standard deviation of 5. At the .05 significance level, is there a difference between the population means?

8. The null and alternate hypotheses are:
- $$H_0: \mu_1 = \mu_2$$
$$H_1: \mu_1 \neq \mu_2$$

A random sample of 15 observations from the first population revealed a sample mean of 350 and a sample standard deviation of 12. A random sample of 17 observations from the second population revealed a sample mean of 342 and a sample standard deviation of 15. At the .10 significance level, is there a difference in the population means?

Note: Use the six-step hypothesis testing procedure for the following exercises.

9. **FILE** Listed below are the 25 players on the opening-day roster of the 2016 New York Yankees Major League Baseball team, their salaries, and fielding positions.

Player	Position	Salary (US\$)
C.C. Sabathia	Starting Pitcher	\$25,000,000
Mark Teixeira	First Base	\$23,125,000
Masahiro Tanaka	Starting Pitcher	\$22,000,000
Jacoby Ellsbury	Center Field	\$21,142,857
Alex Rodriguez	Designated Hitter	\$21,000,000
Brian McCann	Catcher	\$17,000,000
Carlos Beltran	Right Field	\$15,000,000
Brett Gardner	Left Field	\$13,500,000
Chase Headley	Third Base	\$13,000,000
Andrew Miller	Relief Pitcher	\$ 9,000,000
Starlin Castro	Second Base	\$ 7,857,142
Nathan Eovaldi	Starting Pitcher	\$ 5,600,000
Michael Pineda	Starting Pitcher	\$ 4,300,000
Ivan Nova	Relief Pitcher	\$ 4,100,000
Dustin Ackley	Left Field	\$ 3,200,000
Didi Gregorius	Shortstop	\$ 2,425,000
Aaron Hicks	Center Field	\$ 574,000
Austin Romine	Catcher	\$ 556,000
Chasen Shreve	Relief Pitcher	\$ 533,400
Luis Severino	Starting Pitcher	\$ 521,300
Kirby Yates	Relief Pitcher	\$ 511,900
Ronald Torreyes	Second Base	\$ 508,600
Johnny Barbato	Relief Pitcher	\$ 507,500
Dellin Betances	Relief Pitcher	\$ 507,500
Luis Cessa	Relief Pitcher	\$ 507,500

Sort the players into two groups, all pitchers (relief and starting) and position players (all others). Assume equal population standard deviations for the pitchers and the position players. Test the hypothesis that mean salaries of pitchers and position players are equal using the .01 significance level.

10. A recent study compared the time spent together by single- and dual-earner couples. According to the records kept by the wives during the study, the mean amount of time spent together watching television among the single-earner couples was 61 minutes per day, with a standard deviation of 15.5 minutes. For the dual-earner couples, the mean number of minutes spent watching television was 48.4 minutes, with a standard deviation of 18.1 minutes. At the .01 significance level, can we conclude that the single-earner couples on average spend more time watching television together? There were 15 single-earner and 12 dual-earner couples studied.

11. **FILE** Ms. Lisa Monnin is the budget director for Nexus Media Inc. She would like to compare the daily travel expenses for the sales staff and the audit staff. She collected the following sample information.

Sales (\$)	131	135	146	165	136	142	
Audit (\$)	130	102	129	143	149	120	139

At the .10 significance level, can she conclude that the mean daily expenses are greater for the sales staff than the audit staff? What is the *p*-value?

12. **FILE** The Tampa Bay (Florida) Area Chamber of Commerce wanted to know whether the mean weekly salary of nurses was larger than that of school teachers. To investigate, they collected the following information on the amounts earned last week by a sample of school teachers and a sample of nurses.

School Teachers (\$)	1,095	1,076	1,077	1,125	1,034	1,059	1,052	1,070	1,079	1,080	1,092	1,082
Nurses (\$)	1,091	1,140	1,071	1,021	1,100	1,109	1,075	1,079				

Is it reasonable to conclude that the mean weekly salary of nurses is higher? Use the .01 significance level. What is the *p*-value?

Unequal Population Standard Deviations

In the previous sections, it was necessary to assume that the populations had equal standard deviations. To put it another way, we did not know the population standard deviations, but we assumed they were equal. In many cases, this is a reasonable assumption, but what if it is not? In the next chapter, we present a formal method to test the assumption of equal variances. If the variances are not equal, we describe a test of hypothesis that does not require either the equal variance or the normality assumption in Chapter 16.

If it is not reasonable to assume the population standard deviations are equal, then we use a statistic very much like formula (11–2). The sample standard deviations, *s*₁ and *s*₂, are used in place of the respective population standard deviations. In addition, the degrees of freedom are adjusted downward by a rather complex approximation formula. The effect is to reduce the number of degrees of freedom in the test, which will require a larger value of the test statistic to reject the null hypothesis.

The formula for the *t* statistic is:

TEST STATISTIC FOR NO DIFFERENCE
IN MEANS, UNEQUAL VARIANCES

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

(11–5)

The degrees of freedom statistic is found by:

DEGREES OF FREEDOM FOR
UNEQUAL VARIANCE TEST

$$df = \frac{[(s_1^2/n_1) + (s_2^2/n_2)]^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

(11–6)

where *n*₁ and *n*₂ are the respective sample sizes and *s*₁ and *s*₂ are the respective sample standard deviations. If necessary, this fraction is rounded down to an integer value. An example will explain the details.

EXAMPLE

Personnel in a consumer testing laboratory are evaluating the absorbency of paper towels. They wish to compare a set of store brand towels to a similar group of name brand ones. For each brand they dip a ply of the paper into a tub of fluid, allow the paper to drain back into the vat for 2 minutes, and then evaluate the amount of liquid the paper has taken up from the vat. A random sample of nine store brand paper towels absorbed the following amounts of liquid in milliliters.

8 8 3 1 9 7 5 5 12

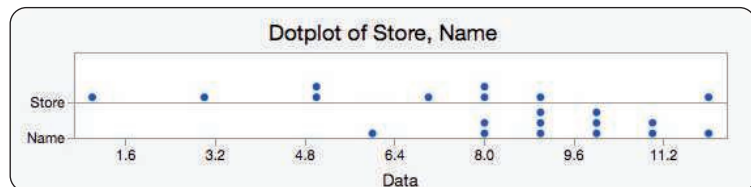
An independent random sample of 12 name brand towels absorbed the following amounts of liquid in milliliters:

12 11 10 6 8 9 9 10 11 9 8 10

Use the .10 significance level and test if there is a difference in the mean amount of liquid absorbed by the two types of paper towels.

SOLUTION

To begin, let's assume that the amounts of liquid absorbed follow the normal probability distribution for both the store brand and the name brand towels. We do not know either of the population standard deviations, so we are going to use the t distribution as the test statistic. The assumption of equal population standard deviations does not appear reasonable. The amount of absorption in the store brand ranges from 1 ml to 12 ml. For the name brand, the amount of absorption ranges from 6 ml to 12 ml. That is, there is considerably more variation in the amount of absorption in the store brand than in the name brand. We observe the difference in the variation in the following dot plot provided by Minitab. The software commands to create a Minitab dot plot are given in Appendix C, Chapter 4, 4-1.



So we decide to use the t distribution and assume that the population standard deviations are not the same.

In the six-step hypothesis testing procedure, the first step is to state the null hypothesis and the alternate hypothesis. The null hypothesis is that there is no difference in the mean amount of liquid absorbed between the two types of paper towels. The alternate hypothesis is that there is a difference.

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

The significance level is .10 and the test statistic follows the t distribution. Because we do not wish to assume equal population standard deviations, we adjust the degrees of freedom using formula (11-6). To do so, we need to find the sample standard deviations. We can use statistical software to quickly find these results.

The respective sample sizes are $n_1 = 9$ and $n_2 = 12$ and the respective standard deviations are 3.321 ml and 1.621 ml.

Variable	<i>n</i>	Mean	Standard Deviation
Store	9	6.444	3.321
Name	12	9.417	1.621

Inserting this information into formula (11–6):

$$df = \frac{\frac{[(s_1^2/n_1) + (s_2^2/n_2)]^2}{(s_1^2/n_1)^2 + (s_2^2/n_2)^2}}{n_1 - 1 + n_2 - 1} = \frac{\frac{[(3.321^2/9) + (1.621^2/12)]^2}{(3.321^2/9)^2 + (1.621^2/12)^2}}{9 - 1 + 12 - 1} = \frac{1.4444^2}{.1877 + .0044} = 10.86$$

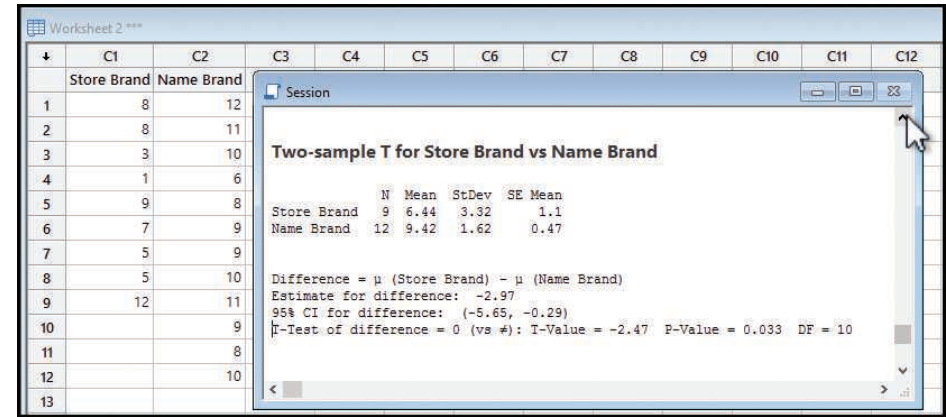
The usual practice is to round down to the integer, so we use 10 degrees of freedom. From Appendix B.5 with 10 degrees of freedom, a two-tailed test, and the .10 significance level, the critical *t* values are −1.812 and 1.812. Our decision rule is to reject the null hypothesis if the computed value of *t* is less than −1.812 or greater than 1.812.

To find the value of the test statistic, we use formula (11–5). Recall that the mean amount of absorption for the store paper towels is 6.444 ml and 9.417 ml for the brand.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{6.444 - 9.417}{\sqrt{\frac{3.321^2}{9} + \frac{1.621^2}{12}}} = -2.474$$

The computed value of *t* is less than the lower critical value, so our decision is to reject the null hypothesis. We conclude that the mean absorption rate for the two towels is not the same.

For this analysis there are many calculations. Statistical software often provides an option to compare two population means with different standard deviations. The Minitab output for this example follows.



SELF-REVIEW 11–3



It is often useful for companies to know who their customers are and how they became customers. A credit card company is interested in whether the owner of the card applied for the card on his or her own or was contacted by a telemarketer. The company obtained the following sample information regarding end-of-the-month balances for the two groups.

Source	Sample Size	Mean	Standard Deviation
Applied	10	\$1,568	\$356
Contacted	8	1,967	857

Is it reasonable to conclude the mean balance is larger for the credit card holders that were contacted by telemarketers than for those who applied on their own for the card? Assume the population standard deviations are not the same. Use the .05 significance level.

- State the null hypothesis and the alternate hypothesis.
- How many degrees of freedom are there?
- What is the decision rule?
- What is the value of the test statistic?
- What is your decision regarding the null hypothesis?
- Interpret the result.

EXERCISES

For exercises 13 and 14, assume the sample populations do not have equal standard deviations and use the .05 significance level: (a) determine the number of degrees of freedom, (b) state the decision rule, (c) compute the value of the test statistic, and (d) state your decision about the null hypothesis.

13. The null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

A random sample of 15 items from the first population showed a mean of 50 and a standard deviation of 5. A sample of 12 items for the second population showed a mean of 46 and a standard deviation of 15.

14. The null and alternate hypotheses are:

$$H_0: \mu_1 \leq \mu_2$$

$$H_1: \mu_1 > \mu_2$$

A random sample of 20 items from the first population showed a mean of 100 and a standard deviation of 15. A sample of 16 items for the second population showed a mean of 94 and a standard deviation of 8. Use the .05 significance level.

15. A recent survey compared the costs of adoption through public and private agencies. For a sample of 16 adoptions through a public agency, the mean cost was \$21,045, with a standard deviation of \$835. For a sample of 18 adoptions through a private agency, the mean cost was \$22,840, with a standard deviation of \$1,545. Can we conclude the mean cost is larger for adopting children through a private agency? Use the .05 significance level.
16. **FILE** Suppose you are an expert on the fashion industry and wish to gather information to compare the amount earned per month by models featuring Liz Claiborne attire with those of Calvin Klein. The following is the amount (\$000) earned per month by a sample of 15 Claiborne models:

\$5.0	\$4.5	\$3.4	\$3.4	\$6.0	\$3.3	\$4.5	\$4.6	\$3.5	\$5.2
4.8	4.4	4.6	3.6	5.0					

The following is the amount (\$000) earned by a sample of 12 Klein models.

\$3.1	\$3.7	\$3.6	\$4.0	\$3.8	\$3.8	\$5.9	\$4.9	\$3.6	\$3.6
2.3	4.0								

Is it reasonable to conclude that Claiborne models earn more? Use the .05 significance level and assume the population standard deviations are not the same.

LO11-3

Test a hypothesis about the mean population difference between paired or dependent observations.

TWO-SAMPLE TESTS OF HYPOTHESIS: DEPENDENT SAMPLES

In the Owens Lawn Care example/solution on page 361, we tested the difference between the means from two independent populations. We compared the mean time required to mount an engine using the Welles method to the time to mount the engine using the Atkins method. The samples were *independent*, meaning that the sample of assembly times using the Welles method was in no way related to the sample of assembly times using the Atkins method.

There are situations, however, in which the samples are not independent. To put it another way, the samples are *dependent* or *related*. As an example, Nickel Savings and Loan employs two firms, Schadek Appraisals and Bowyer Real Estate, to appraise the value of the real estate properties on which it makes loans. It is important that these two firms be similar in their appraisal values. To review the consistency of the two appraisal firms, Nickel Savings randomly selects 10 homes and has both Schadek Appraisals and Bowyer Real Estate appraise the values of the selected homes. For each home, there will be a pair of appraisal values. That is, for each home there will be an appraised value from both Schadek Appraisals and Bowyer Real Estate. The appraised values depend on, or are related to, the home selected. This is also referred to as a **paired sample**.



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For hypothesis testing, we are interested in the distribution of the *differences* in the appraised value of each home. Hence, there is only one sample. To put it more formally, we are investigating whether the mean of the distribution of differences in the appraised values is 0. The sample is made up of the *differences* between the appraised values determined by Schadek Appraisals and the values from Bowyer Real Estate. If the two appraisal firms are reporting similar estimates, then sometimes Schadek Appraisals will be the higher value and sometimes Bowyer Real Estate will have the higher value. However, the mean of the distribution of differences will be 0. On the other hand, if one of the

firms consistently reports larger appraisal values, then the mean of the distribution of the differences will not be 0.

We will use the symbol μ_d to indicate the population mean of the distribution of differences. We assume the distribution of the population of differences is approximately normally distributed. The test statistic follows the t distribution and we calculate its value from the following formula:

PAIRED t TEST

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \quad (11-7)$$

There are $n - 1$ degrees of freedom and

$\bar{d}$ is the mean of the difference between the paired or related observations.

s_d is the standard deviation of the differences between the paired or related observations.

n is the number of paired observations.

The standard deviation of the differences is computed by the familiar formula for the standard deviation [see formula (3-9)], except d is substituted for x . The formula is:

$$s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}}$$

The following example illustrates this test.

EXAMPLE

Recall that Nickel Savings and Loan wishes to compare the two companies it uses to appraise the value of residential homes. Nickel Savings selected a sample of 10 residential properties and scheduled both firms for an appraisal. The results, reported in \$000, are:

Home	Schadek	Bowyer
A	235	228
B	210	205
C	231	219
D	242	240
E	205	198
F	230	223
G	231	227
H	210	215
I	225	222
J	249	245

At the .05 significance level, can we conclude there is a difference between the firms' mean appraised home values?

SOLUTION

The first step is to state the null and the alternate hypotheses. In this case, a two-tailed alternative is appropriate because we are interested in determining whether there is a *difference* in the firms' appraised values. We are not interested in showing whether one particular firm appraises property at a higher value than the other. The question is whether the sample differences in the appraised values could have come from a population with a mean of 0. If the population mean of the differences is 0, then we conclude that there is no difference between the two firms' appraised values. The null and alternate hypotheses are:

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

There are 10 homes appraised by both firms, so $n = 10$, and $df = n - 1 = 10 - 1 = 9$. We have a two-tailed test, and the significance level is .05. To determine the critical value, go to Appendix B.5 and move across the row with 9 degrees of freedom to the column for a two-tailed test and the .05 significance level. The value at the intersection is 2.262. This value appears in the box in Table 11-2. The decision rule is to reject the null hypothesis if the computed value of t is less than -2.262 or greater than 2.262 . Here are the computational details.

Home	Schadek	Bowyer	Difference, d	$(d - \bar{d})$	$(d - \bar{d})^2$
A	235	228	7	2.4	5.76
B	210	205	5	0.4	0.16
C	231	219	12	7.4	54.76
D	242	240	2	-2.6	6.76
E	205	198	7	2.4	5.76
F	230	223	7	2.4	5.76
G	231	227	4	-0.6	0.36
H	210	215	-5	-9.6	92.16
I	225	222	3	-1.6	2.56
J	249	245	4	-0.6	0.36
			46	0	174.40

$$\bar{d} = \frac{\sum d}{n} = \frac{46}{10} = 4.60$$
$$s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}} = \sqrt{\frac{174.4}{10 - 1}} = 4.402$$

Using formula (11–7), the value of the test statistic is 3.305, found by

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} = \frac{4.6}{4.402 / \sqrt{10}} = \frac{4.6}{1.3920} = 3.305$$

Because the computed *t* falls in the rejection region, the null hypothesis is rejected. The population distribution of differences does not have a mean of 0. We conclude that there is a difference between the firms’ mean appraised home values. The largest difference of \$12,000 is for Home 3. Perhaps that would be an appropriate place to begin a more detailed review.

To find the *p*-value, we use Appendix B.5 and the section for a two-tailed test. Move along the row with 9 degrees of freedom and find the values of *t* that are closest to our calculated value. For a .01 significance level, the value of *t* is 3.250. The computed value is larger than this value, but smaller than the value of 4.781 corresponding to the .001 significance level. Hence, the *p*-value is less than .01. This information is highlighted in Table 11–2.

TABLE 11–2 A Portion of the *t* Distribution from Appendix B.5

Confidence Intervals						
	80%	90%	95%	98%	99%	<i>p</i> -value between 0.01 and 0.001
<i>df</i>	Level of Significance for One-Tailed Test					
	0.10	0.05	0.025	0.01	0.005	
	Level of Significance for Two-Tailed Test					
	0.20	0.10	0.05	0.02	0.01	0.001
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.924
Critical <i>t</i> -statistic for 0.05		2.132	2.776	3.747	4.604	8.610
		2.015	2.571	3.365	4.032	6.869
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.408
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587

Excel’s statistical analysis software has a procedure called “t-Test: Paired Two-Sample for Means” that will perform the calculations of formula (11–7). The output from this procedure is given below.

The computed value of *t* is 3.305, and the two-tailed *p*-value is .009. Because the *p*-value is less than .05, we reject the hypothesis that the mean of the distribution of the differences between the appraised values is zero. In fact, this *p*-value is between .01 and .001. There is a small likelihood that the null hypothesis is true.

	A	B	C	D	E	F	G
1	Home	Schadek	Bowyer		t-Test: Paired Two-Sample for Means		
2	A	235	228				
3	B	210	205			Schadek	Bowyer
4	C	231	219		Mean	226.800	222.200
5	D	242	240		Variance	208.844	204.178
6	E	205	198		Observations	10.000	10.000
7	F	230	223		Pearson Correlation	0.953	
8	G	231	227		Hypothesized Mean Difference	0.000	
9	H	210	215		df	9.000	
10	I	225	222		t Stat	3.305	
11	J	249	245		P(T<=t) one-tail	0.005	
12					t Critical one-tail	1.833	
13					P(T<=t) two-tail	0.009	
14					t Critical two-tail	2.262	
15							

LO11-4

Explain the difference between dependent and independent samples.

COMPARING DEPENDENT AND INDEPENDENT SAMPLES

Beginning students are often confused by the difference between tests for independent samples [formula (11–4)] and tests for dependent samples [formula (11–7)]. How do we tell the difference between dependent and independent samples? There are two types of dependent samples: (1) those characterized by a measurement, an intervention of some type, and then another measurement; and (2) a matching or pairing of the observations. To explain further:

1. The first type of dependent sample is characterized by a measurement followed by an intervention of some kind and then another measurement. This could be called a “before” and “after” study. Two examples will help to clarify. Suppose we want to show that, by placing speakers in the production area and playing soothing music, we are able to increase production. We begin by selecting a sample of workers and measuring their output under the current conditions. The speakers are then installed in the production area, and we again measure the output of the same workers. There are two measurements, before placing the speakers in the production area and after. The intervention is placing speakers in the production area.

A second example involves an educational firm that offers courses designed to increase test scores and reading ability. Suppose the firm wants to offer a course that will help high school juniors increase their SAT scores. To begin, each student takes the SAT in the junior year in high school. During the summer between the junior and senior year, they participate in the course that gives them tips on taking tests. Finally, during the fall of their senior year in high school, they retake the SAT. Again, the procedure is characterized by a measurement (taking the SAT as a junior), an intervention (the summer workshops), and another measurement (taking the SAT during their senior year).

2. The second type of dependent sample is characterized by matching or pairing observations. The previous example/solution regarding Nickel Savings illustrates dependent samples. A property is selected and both firms appraise the same property. As a second example, suppose an industrial psychologist wishes to study the intellectual similarities of newly married couples. She selects a sample of newlyweds. Next, she administers a standard intelligence test to both the man and woman to determine the difference in the scores. Notice the matching that occurred: comparing the scores that are paired or matched by marriage.

Why do we prefer dependent samples to independent samples? By using dependent samples, we are able to reduce the variation in the sampling distribution. To illustrate, we will use the Nickel Savings and Loan example/solution just completed. Suppose we assume that we have two independent samples of real estate property for appraisal and conduct the following test of hypothesis, using formula (11–4). The null and alternate hypotheses are:

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

There are now two independent samples of 10 each. So the number of degrees of freedom is $10 + 10 - 2 = 18$. From Appendix B.5, for the .05 significance level, H_0 is rejected if t is less than -2.101 or greater than 2.101 .

We use Excel to find the means and standard deviations of the two independent samples as shown in the Chapter 3 section of Appendix C. The Excel instructions to find the pooled variance and the value of the “t Stat” are in the Chapter 11 section in Appendix C. These values are highlighted in yellow.

	A	B	C	D	E	F	G	H
1	Home	Schadek	Bowyer		t-Test: Two-Sample Assuming Equal Variances			
2	A	235	228					
3	B	210	205			Schadek	Bowyer	
4	C	231	219		Mean	226.800	222.200	
5	D	242	240		Variance	208.844	204.178	
6	E	205	198		Observations	10.000	10.000	
7	F	230	223		Pooled Variance	206.511		
8	G	231	227		Hypothesized Mean Difference	0.000		
9	H	210	215		df	18.000		
10	I	225	222		t Stat	0.716		
11	J	249	245		P(T<=t) one-tail	0.242		
12					t Critical one-tail	1.734		
13	Mean =	226.80	222.20		P(T<=t) two-tail	0.483		
14	S =	14.45	14.29		t Critical two-tail	2.101		
15								

The mean of the appraised value of the 10 properties by Schadek is \$226,800, and the standard deviation is \$14,500. For Bowyer Real Estate, the mean appraised value is \$222,200, and the standard deviation is \$14,290. To make the calculations easier, we use \$000 instead of \$. The value of the pooled estimate of the variance from formula (11–3) is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(10 - 1)(14.45^2) + (10 - 1)(14.29)^2}{10 + 10 - 2} = 206.50$$

From formula (11–4), t is 0.716.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{226.8 - 222.2}{\sqrt{206.50 \left(\frac{1}{10} + \frac{1}{10} \right)}} = \frac{4.6}{6.4265} = 0.716$$

The computed t (0.716) is less than 2.101, so the null hypothesis is not rejected. We cannot show that there is a difference in the mean appraisal value. That is not the same conclusion that we got before! Why does this happen? The numerator is the same in the paired observations test (4.6). However, the denominator is smaller. In the paired test, the denominator is 1.3920 (see the calculations on page 372 in the previous section). In the case of the independent samples, the denominator is 6.4265. There is more variation or uncertainty. This accounts for the difference in the t values and the difference in the

statistical decisions. The denominator measures the standard error of the statistic. When the samples are *not* paired, two kinds of variation are present: differences between the two appraisal firms and the difference in the value of the real estate. Properties numbered 4 and 10 have relatively high values, whereas number 5 is relatively low. These data show how different the values of the property are, but we are really interested in the difference between the two appraisal firms.

In sum, when we can pair or match observations that measure differences for a common variable, a hypothesis test based on dependent samples is more sensitive to detecting a significant difference than a hypothesis test based on independent samples. In the case of comparing the property valuations by Schadek Appraisals and Bowyer Real Estate, the hypothesis test based on dependent samples eliminates the variation between the values of the properties and focuses only on the comparisons in the two appraisals for each property. There is a bit of bad news here. In the dependent samples test, the degrees of freedom are half of what they are if the samples are not paired. For the real estate example, the degrees of freedom drop from 18 to 9 when the observations are paired. However, in most cases, this is a small price to pay for a better test.

SELF-REVIEW 11-4



Advertisements by Core Fitness Center claim that completing its course will result in losing weight. A random sample of eight recent participants showed the following weights before and after completing the course. At the .01 significance level, can we conclude the students lost weight?

Name	Before	After
Hunter	155	154
Cashman	228	207
Mervine	141	147
Massa	162	157
Creola	211	196
Peterson	164	150
Redding	184	170
Poust	172	165

- State the null hypothesis and the alternate hypothesis.
- What is the critical value of t ?
- What is the computed value of t ?
- Interpret the result. What is the p -value?
- What assumption needs to be made about the distribution of the differences?

EXERCISES

17. The null and alternate hypotheses are:

$$H_0: \mu_d \leq 0$$

$$H_1: \mu_d > 0$$

The following sample information shows the number of defective units produced on the day shift and the afternoon shift for a sample of four days last month.

	Day			
	1	2	3	4
Day shift	10	12	15	19
Afternoon shift	8	9	12	15

At the .05 significance level, can we conclude there are more defects produced on the day shift?

18. The null and alternate hypotheses are:

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

The following paired observations show the number of traffic citations given for speeding by Officer Dhondt and Officer Meredith of the South Carolina Highway Patrol for the last five months.

	Number of Citations Issued				
	May	June	July	August	September
Officer Dhondt	30	22	25	19	26
Officer Meredith	26	19	20	15	19

At the .05 significance level, is there a difference in the mean number of citations given by the two officers?

Note: Use the six-step hypothesis testing procedure to solve the following exercises.

19. **FILE** The management of Discount Furniture, a chain of discount furniture stores in the Northeast, designed an incentive plan for salespeople. To evaluate this innovative plan, 12 salespeople were selected at random, and their weekly incomes before and after the plan were recorded.

Salesperson	Before	After
Sid Mahone	\$320	\$340
Carol Quick	290	285
Tom Jackson	421	475
Andy Jones	510	510
Jean Sloan	210	210
Jack Walker	402	500
Peg Mancuso	625	631
Anita Loma	560	560
John Cuso	360	365
Carl Utz	431	431
A. S. Kushner	506	525
Fern Lawton	505	619

Was there a significant increase in the typical salesperson’s weekly income due to the innovative incentive plan? Use the .05 significance level. Estimate the *p*-value, and interpret it.

20. **FILE** The federal government recently granted funds for a special program designed to reduce crime in high-crime areas. A study of the results of the program in eight high-crime areas of Miami, Florida, yielded the following results.

	Number of Crimes by Area							
	A	B	C	D	E	F	G	H
Before	14	7	4	5	17	12	8	9
After	2	7	3	6	8	13	3	5

Has there been a decrease in the number of crimes since the inauguration of the program? Use the .01 significance level. Estimate the *p*-value.

CHAPTER SUMMARY

- I. In comparing two population means, we wish to know whether they could be equal.
- A. We are investigating whether the distribution of the difference between the means could have a mean of 0.
 - B. The test statistic follows the standard normal distribution if the population standard deviations are known.
 - 1. The two populations follow normal distributions.
 - 2. The samples are from independent populations.
 - 3. The formula to compute the value of z is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad (11-2)$$

- II. The test statistic to compare two means is the t distribution if the population standard deviations are not known.
- A. Both populations are approximately normally distributed.
 - B. The populations must have equal standard deviations.
 - C. The samples are independent.
 - D. Finding the value of t requires two steps.
 - 1. The first step is to pool the standard deviations according to the following formula:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \quad (11-3)$$

- 2. The value of t is computed from the following formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (11-4)$$

- 3. The degrees of freedom for the test are $n_1 + n_2 - 2$.
- III. If we cannot assume the population standard deviations are equal, we adjust the degrees of freedom and the formula for finding t .
- A. We determine the degrees of freedom based on the following formula.

$$df = \frac{[(s_1^2/n_1) + (s_2^2/n_2)]^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \quad (11-6)$$

- B. The value of the test statistic is computed from the following formula.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (11-5)$$

- IV. For dependent samples, we assume the population distribution of the paired differences has a mean of 0.
- A. We first compute the mean and the standard deviation of the sample differences.
 - B. The value of the test statistic is computed from the following formula:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \quad (11-7)$$

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
s_p^2	Pooled sample variance	<i>s squared sub p</i>
$\bar{x}_1$	Mean of the first sample	<i>x bar sub 1</i>
$\bar{x}_2$	Mean of the second sample	<i>x bar sub 2</i>
$\bar{d}$	Mean of the difference between dependent observations	<i>d bar</i>
s_d	Standard deviation of the difference between dependent observations	<i>s sub d</i>

CHAPTER EXERCISES

21. A recent study focused on the number of times men and women who live alone buy take-out dinner in a month. Assume that the distributions follow the normal probability distribution and the population standard deviations are equal. The information is summarized below.

Statistic	Men	Women
Sample mean	24.51	22.69
Sample standard deviation	4.48	3.86
Sample size	35	40

- At the .01 significance level, is there a difference in the mean number of times men and women order take-out dinners in a month? What is the p -value?
22. Clark Heter is an industrial engineer at Lyons Products. He would like to determine whether there are more units produced on the night shift than on the day shift. The mean number of units produced by a sample of 54 day-shift workers was 345. The mean number of units produced by a sample of 60 night-shift workers was 351. Assume the population standard deviation of the number of units produced on the day shift is 21 and 28 on the night shift. Using the .05 significance level, is the number of units produced on the night shift larger?
23. Fry Brothers Heating and Air Conditioning Inc. employs Larry Clark and George Murnen to make service calls to repair furnaces and air-conditioning units in homes. Tom Fry, the owner, would like to know whether there is a difference in the mean number of service calls they make per day. A random sample of 40 days last year showed that Larry Clark made an average of 4.77 calls per day. For a sample of 50 days George Murnen made an average of 5.02 calls per day. Assume the population standard deviation for Larry Clark is 1.05 calls per day and 1.23 calls per day for George Murnen. At the .05 significance level, is there a difference in the mean number of calls per day between the two employees? What is the p -value?
24. A coffee manufacturer is interested in whether the mean daily consumption of regular-coffee drinkers is less than that of decaffeinated-coffee drinkers. Assume the population standard deviation for those drinking regular coffee is 1.20 cups per day and 1.36 cups per day for those drinking decaffeinated coffee. A random sample of 50 regular-coffee drinkers showed a mean of 4.35 cups per day. A sample of 40 decaffeinated-coffee drinkers showed a mean of 5.84 cups per day. Use the .01 significance level. Compute the p -value.
25. A cell phone company offers two plans to its subscribers. At the time new subscribers sign up, they are asked to provide some demographic information. The mean

- yearly income for a sample of 40 subscribers to Plan A is \$57,000 with a standard deviation of \$9,200. For a sample of 30 subscribers to Plan B, the mean income is \$61,000 with a standard deviation of \$7,100. At the .05 significance level, is it reasonable to conclude the mean income of those selecting Plan B is larger? What is the p -value?
26. A computer manufacturer offers technical support that is available 24 hours a day, 7 days a week. Timely resolution of these calls is important to the company's image. For 35 calls that were related to software, technicians resolved the issues in a mean time of 18 minutes with a standard deviation of 4.2 minutes. For 45 calls related to hardware, technicians resolved the problems in a mean time of 15.5 minutes with a standard deviation of 3.9 minutes. At the .05 significance level, does it take longer to resolve software issues? What is the p -value?
27. Music streaming services are the most popular way to listen to music. Data gathered over the last 12 months show Apple Music was used by an average of 1.65 million households with a sample standard deviation of 0.56 million family units. Over the same 12 months Spotify was used by an average of 2.2 million families with a sample standard deviation of 0.30 million. Assume the population standard deviations are not the same. Using a significance level of .05, test the hypothesis of no difference in the mean number of households picking either service.
28. Businesses such as General Mills, Kellogg's, and Betty Crocker regularly use coupons to build brand allegiance and stimulate sales. Marketers believe that the users of paper coupons are different from the users of e-coupons accessed through the Internet. One survey recorded the age of each person who redeemed a coupon along with the type of coupon (either paper or electronic). The sample of 25 traditional paper-coupon clippers had a mean age of 39.5 with a standard deviation of 4.8. The sample of 35 e-coupon users had a mean age of 33.6 years with a standard deviation of 10.9. Assume the population standard deviations are not the same. Using a significance level of .01, test the hypothesis of no difference in the mean ages of the two groups of coupon clients.
29. The owner of Bun 'N' Run Hamburgers wishes to compare the sales per day at two locations. The mean number sold for 10 randomly selected days at the Northside site was 83.55, and the standard deviation was 10.50. For a random sample of 12 days at the Southside location, the mean number sold was 78.80 and the standard deviation was 14.25. At the .05 significance level, is there a difference in the mean number of hamburgers sold at the two locations? What is the p -value?
30. **FILE** Educational Technology, Inc. sells software to provide guided homework problems for a statistics course. They would like to know if students who use the software score better on exams. A sample of students who used the software had the following exam scores: 86, 78, 66, 83, 84, 81, 84, 109, 65, and 102. Students who did not use the software had the following exam scores: 91, 71, 75, 76, 87, 79, 73, 76, 79, 78, 87, 90, 76, and 72. Assume the population standard deviations are not the same. At the .10 significance level, can we conclude that there is a difference in the mean exam scores for the two groups of students?
31. **FILE** The Willow Run Outlet Mall has two Haggar Outlet Stores, one located on Peach Street and the other on Plum Street. The two stores are laid out differently, but both store managers claim their layout maximizes the amounts customers will purchase on impulse. A sample of 10 customers at the Peach Street store revealed they spent the following amounts on impulse purchases: \$17.58, \$19.73, \$12.61, \$17.79, \$16.22, \$15.82, \$15.40, \$15.86, \$11.82, and \$15.85. A sample of 14 customers at the Plum Street store revealed they spent the following amounts on impulse purchases: \$18.19, \$20.22, \$17.38, \$17.96, \$23.92, \$15.87, \$16.47, \$15.96, \$16.79, \$16.74, \$21.40, \$20.57, \$19.79, and \$14.83. At the .01 significance level, is there a difference in the mean amounts purchased on impulse at the two stores?
32. **FILE** Grand Strand Family Medical Center treats minor medical emergencies for visitors to the Myrtle Beach area. There are two facilities, one in the Little River Area and the other in Murrells Inlet. The Quality Assurance Department wishes to compare the mean

waiting time for patients at the two locations. Samples of the waiting times for each location, reported in minutes, follow:

Location	Waiting Time											
Little River	31	28	29	22	29	18	32	25	29	26		
Murrells Inlet	22	23	26	27	26	25	30	29	23	23	27	22

Assume the population standard deviations are not the same. At the .05 significance level, is there a difference in the mean waiting time?

33. **FILE** Commercial Bank and Trust Company is studying the use of its automatic teller machines (ATMs). Of particular interest is whether young adults (under 25 years) use the machines more than senior citizens. To investigate further, samples of customers under 25 years of age and customers over 60 years of age were selected. The number of ATM transactions last month was determined for each selected individual, and the results are shown below. At the .01 significance level, can bank management conclude that younger customers use the ATMs more?

Under 25	10	10	11	15	7	11	10	9				
Over 60	4	8	7	7	4	5	1	7	4	10	5	

34. **FILE** Two of the teams competing in the *America's Cup* race are Team Oracle U.S.A. and Land Rover BAR. They race their boats over a part of the course several times. Below are a sample of times in minutes for each boat. Assume the population standard deviations are not the same. At the .05 significance level, can we conclude that there is a difference in their mean times?

Boat	Time (minutes)											
Land Rover BAR	12.9	12.5	11.0	13.3	11.2	11.4	11.6	12.3	14.2	11.3		
Team Oracle	14.1	14.1	14.2	17.4	15.8	16.7	16.1	13.3	13.4	13.6	10.8	19.0

35. **FILE** The manufacturer of an MP3 player wanted to know whether a 10% reduction in price is enough to increase the sales of its product. To investigate, the owner randomly selected eight outlets and sold the MP3 player at the reduced price. At seven randomly selected outlets, the MP3 player was sold at the regular price. Reported below is the number of units sold last month at the regular and reduced prices at the randomly selected outlets. At the .01 significance level, can the manufacturer conclude that the price reduction resulted in an increase in sales?

Regular price	138	121	88	115	141	125	96		
Reduced price	128	134	152	135	114	106	112	120	

36. **FILE** A number of minor automobile accidents occur at various high-risk intersections in Teton County despite traffic lights. The Traffic Department claims that a modification in the type of light will reduce these accidents. The county commissioners have agreed to a proposed experiment. Eight intersections were chosen at random, and the lights at those intersections were modified. The numbers of minor accidents during a six-month period before and after the modifications were:

	Number of Accidents							
	A	B	C	D	E	F	G	H
Before modification	5	7	6	4	8	9	8	10
After modification	3	7	7	0	4	6	8	2

At the .01 significance level, is it reasonable to conclude that the modification reduced the number of traffic accidents?

37. **FILE** Lester Hollar is vice president for human resources for a large manufacturing company. In recent years, he has noticed an increase in absenteeism that he thinks is related to the general health of the employees. Four years ago, in an attempt to improve the situation, he began a fitness program in which employees exercise during their lunch hour. To evaluate the program, he selected a random sample of eight participants and found the number of days each was absent in the six months before the exercise program began and in the six months following the exercise program. Below are the results. At the .05 significance level, can he conclude that the number of absences has declined? Estimate the p -value.

Employee	Before	After
Bauman	6	5
Briggs	6	2
Dottellis	7	1
Lee	7	3
Perralt	4	3
Rielly	3	6
Steinmetz	5	3
Stoltz	6	7

38. **FILE** The president of the American Insurance Institute wants to compare the yearly costs of auto insurance offered by two leading companies. He selects a sample of 15 families, some with only a single insured driver, others with several teenage drivers, and pays each family a stipend to contact the two companies and ask for a price quote. To make the data comparable, certain features, such as the deductible amount and limits of liability, are standardized. The data for the sample of families and their two insurance quotes are reported below. At the .10 significance level, can we conclude that there is a difference in the amounts quoted?

Family	Midstates Car Insurance	Gecko Mutual Insurance
Becker	\$2,090	\$1,610
Berry	1,683	1,247
Cobb	1,402	2,327
Debuck	1,830	1,367
DuBrul	930	1,461
Eckroate	697	1,789
German	1,741	1,621
Glasson	1,129	1,914
King	1,018	1,956
Kucic	1,881	1,772
Meredith	1,571	1,375
Obeid	874	1,527
Price	1,579	1,767
Phillips	1,577	1,636
Tresize	860	1,188

39. Fairfield Homes is developing two parcels near Pigeon Fork, Tennessee. In order to test different advertising approaches, it uses different media to reach potential

buyers. The mean annual family income for 15 people making inquiries at the first development is \$150,000, with a standard deviation of \$40,000. A corresponding sample of 25 people at the second development had a mean of \$180,000, with a standard deviation of \$30,000. Assume the population standard deviations are the same. At the .05 significance level, can Fairfield conclude that the population means are different?

40. A candy company taste-tested two chocolate bars, one with almonds and one without almonds. A panel of testers rated the bars on a scale of 0 to 5, with 5 indicating the highest taste rating. Assume the population standard deviations are equal. At the .05 significance level, do the ratings show a difference between chocolate bars with or without almonds?

With Almonds	Without Almonds
3	0
1	4
2	4
3	3
1	4
1	
2	

41. **FILE** An investigation of the effectiveness of an antibacterial soap in reducing operating room contamination resulted in the accompanying table. The new soap was tested in a sample of eight operating rooms in the greater Seattle area during the last year. The following table reports the contamination levels before and after the use of the soap for each operating room.

	Operating Room							
	A	B	C	D	E	F	G	H
Before	6.6	6.5	9.0	10.3	11.2	8.1	6.3	11.6
After	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0

At the .05 significance level, can we conclude the contamination measurements are lower after use of the new soap?

42. **FILE** The following data on annual rates of return were collected from eleven randomly selected stocks listed on the New York Stock Exchange (“the big board”) and twelve randomly selected stocks listed on NASDAQ. Assume the population standard deviations are the same. At the .10 significance level, can we conclude that the annual rates of return are higher on the big board?

NYSE	NASDAQ
15.0	8.8
10.7	6.0
20.2	14.4
18.6	19.1
19.1	17.6
8.7	17.8
17.8	15.9
13.8	17.9
22.7	21.6
14.0	6.0
26.1	11.9
	23.4

43. **FILE** The city of Laguna Beach operates two public parking lots. The Ocean Drive parking lot can accommodate up to 125 cars and the Rio Rancho parking lot can accommodate up to 130 cars. City planners are considering increasing the size of the lots and changing the fee structure. To begin, the Planning Office would like some information on the number of cars in the lots at various times of the day. A junior planner officer is assigned the task of visiting the two lots at random times of the day and evening and counting the number of cars in the lots. The study lasted over a period of one month. Below is the number of cars in the lots for 25 visits of the Ocean Drive lot and 28 visits of the Rio Rancho lot. Assume the population standard deviations are equal.

Ocean Drive												
89	115	93	79	113	77	51	75	118	105	106	91	54
63	121	53	81	115	67	53	69	95	121	88	64	
Rio Rancho												
128	110	81	126	82	114	93	40	94	45	84	71	74
92	66	69	100	114	113	107	62	77	80	107	90	129
105	124											

Is it reasonable to conclude that there is a difference in the mean number of cars in the two lots? Use the .05 significance level.

44. **FILE** The amount of income spent on housing is an important component of the cost of living. The total costs of housing for homeowners might include mortgage payments, property taxes, and utility costs (water, heat, electricity). An economist selected a sample of 20 homeowners in New England and then calculated these total housing costs as a percent of monthly income, 5 years ago and now. The information is reported below. Is it reasonable to conclude the percent is less now than 5 years ago?

Homeowner	Five Years Ago	Now	Homeowner	Five Years Ago	Now
Holt	17%	10%	Lozier	35%	32%
Pierse	20	39	Cieslinski	16	32
Merenick	29	37	Rowatti	23	21
Lanoue	43	27	Koppel	33	12
Fagan	36	12	Rumsey	44	40
Bobko	43	41	McGinnis	44	42
Kippert	45	24	Pierce	28	22
San Roman	19	26	Roll	29	19
Kurimsky	49	28	Lang	39	35
Davison	49	26	Miller	22	12

45. **FILE** The CVS Pharmacy located on US 17 in Murrells Inlet has been one of the busiest pharmaceutical retail stores in South Carolina for many years. To try and capture more business in the area, CVS top management opened another store about 6 miles west on SC 707. After a few months, CVS management decided to compare the business volume at the two stores. One way to measure business volume is to count the number of cars in the store parking lots on random days and times. The results of the survey from the last 3 months of the year are reported below. To explain, the first observation was on October 2 at 20:52 military time (8:52 p.m.). At that time there were four cars in the US 17 lot and nine cars in the SC 707 lot. At the .05 significance level, is it reasonable to

conclude that, based on vehicle counts, the US 17 store has more business volume than the SC 707 store?

Date	Time	Vehicle Count	
		US 17	SC 707
Oct 2	20:52	4	9
Oct 11	19:30	5	7
Oct 15	22:08	9	12
Oct 19	11:42	4	5
Oct 25	15:32	10	8
Oct 26	11:02	9	15
Nov 3	11:22	13	7
Nov 5	19:09	20	3
Nov 8	15:10	15	14
Nov 9	13:18	15	11
Nov 15	22:38	13	11
Nov 17	18:46	16	12
Nov 21	15:44	17	8
Nov 22	15:34	15	3
Nov 27	21:42	20	6
Nov 29	9:57	17	13
Nov 30	17:58	5	9
Dec 3	19:54	7	13
Dec 15	18:20	11	6
Dec 16	18:25	14	15
Dec 17	11:08	8	8
Dec 22	21:20	10	3
Dec 24	15:21	4	6
Dec 25	20:21	7	9
Dec 30	14:25	19	4

46. **FILE** A goal of financial literacy for children is to learn how to manage money wisely. One question is: How much money do children have to manage? A recent study by Schnur Educational Research Associates randomly sampled 15 children between 8 and 10 years old and 18 children between 11 and 14 years old and recorded their monthly allowance. Is it reasonable to conclude that the mean allowance received by children between 11 and 14 years is more than the allowance received by children between 8 and 10 years? Use the .01 significance level. What is the *p*-value?

8–10 Years	11–14 Years	8–10 Years	11–14 Years
26	49	26	41
33	44	25	38
30	42	27	44
26	38	29	39
34	39	34	50
26	41	32	49
27	39		41
27	38		42
30	38		30

DATA ANALYTICS

47. **FILE** The North Valley Real Estate data reports information on the homes sold last year.
- At the .05 significance level, can we conclude that there is a difference in the mean selling price of homes with a pool and homes without a pool?
 - At the .05 significance level, can we conclude that there is a difference in the mean selling price of homes with an attached garage and homes without an attached garage?
 - At the .05 significance level, can we conclude that there is a difference in the mean selling price of homes that are in default on the mortgage?
48. **FILE** Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season.
- At the .05 significance level, can we conclude that there is a difference in the mean salary of teams in the American League versus teams in the National League?
 - At the .05 significance level, can we conclude that there is a difference in the mean home attendance of teams in the American League versus teams in the National League?
 - Compute the mean and the standard deviation of the number of wins for the 10 teams with the highest salaries. Do the same for the 10 teams with the lowest salaries. At the .05 significance level, is there a difference in the mean number of wins for the two groups? At the .05 significance level, is there a difference in the mean attendance for the two groups?
49. **FILE** Refer to the Lincolnville School District bus data. Is there a difference in the mean maintenance cost for the diesel versus the gasoline buses? Use the .05 significance level.